

Day-by-Day Breakdown

Day 1: RAG Foundations & Architecture

Goal: Understand the RAG pipeline end-to-end

- What is RAG? Why not just fine-tuning?
 - RAG vs. Fine-tuning vs. Prompt Engineering tradeoffs
 - Core pipeline: Index → Retrieve → Augment → Generate
 - Real-world use cases (Q&A, documentation search, domain experts, etc.)
 - Vector databases: embedding models, similarity search, indexing
 - Hands-on: Index a small document corpus with embeddings (use Chroma/Qdrant)
 - Lab 1: Build a naive RAG pipeline (100 docs, basic retrieval)
 - Reading: "Retrieval-Augmented Generation for Knowledge-Intensive Tasks" (Lewis et al., 2020)
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Day 2: Chunking Strategies & Text Processing

Goal: Master document preparation and chunking

- Why chunking matters (context window, relevance, latency)
 - Chunking strategies:
 - **Fixed-size chunking** (pros: simple; cons: semantic breaks)
 - **Semantic chunking** (pros: coherent; cons: slower)
 - **Recursive chunking** (pros: hierarchical; cons: complex)
 - **Sliding window** (pros: context overlap; cons: redundancy)
 - **Document structure-aware** (headers, tables, code blocks)
 - Hands-on chunking experiments:
 - Implement 3-4 chunking strategies
 - Evaluate chunk coherence & retrieval quality
 - Lab 2: Compare chunking strategies on a real dataset (measure retrieval accuracy)
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Day 3: Advanced Retrieval & Reranking

Goal: Improve retrieval quality beyond simple similarity

- BM25 + hybrid search (dense + sparse)

- Query expansion & rewriting
 - Reranking models (cross-encoders vs. bi-encoders)
 - Metadata filtering & structured retrieval
 - Lab 3: Implement hybrid BM25 + dense retrieval
 - Lab 4: Add a reranker (use Cohere/LiteLLM cross-encoder)
 - Measure: MRR, NDCG, Recall@K
 - Deep Dive: When does reranking help? Cost vs. accuracy.
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Day 4: RAG Evaluation & Metrics

Goal: Learn to measure RAG quality rigorously

- Retrieval metrics: Precision, Recall, NDCG, MRR
 - Generation metrics: BLEU, ROUGE, BERTScore, semantic similarity
 - RAG-specific: Faithfulness, relevance, consistency
 - Evaluation frameworks (RAGAS, TruLens, LlamaIndex eval)
 - Lab 5: Build an evaluation pipeline for your RAG
 - Create ground-truth Q&A pairs from docs
 - Run retrieval + generation evals
 - Identify bottlenecks (is it retrieval or generation?)
 - Tool: Use Langfuse/LlamaIndex for instrumentation
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Day 5: Advanced RAG Architectures & Project

Goal: Explore variants and consolidate learning

- RAG variants recap:
 - **Naive RAG** (your baseline)
 - **Advanced RAG** (better retrieval, reranking, query routing)
 - **Hybrid RAG** (combines retrieval + generation feedback loops)
 - **Agentic RAG** (multi-step reasoning, tool use)
 - **Graph-based RAG** (knowledge graphs, entity relationships)
 - **Multi-modal RAG** (images, tables, code)
- Capstone Lab 6: Build an end-to-end RAG system
 - Pick a domain (legal docs, medical papers, API docs, etc.)
 - Implement chunking strategy of choice

- Add hybrid retrieval + reranking
- Integrate with Claude/LLM
- Evaluate & write a summary report
- Code review & optimization
- Bonus: Deploy to simple API (FastAPI + Qdrant) if time permits

Chunking Strategies - Quick Reference

Strategy	Pros	Cons	Use Case
Fixed-size	Simple, fast	Semantic breaks, rigid	Quick prototyping
Semantic	Coherent chunks	Slow, expensive	Precision-critical tasks
Recursive	Hierarchical context	Complex implementation	Long documents
Sliding Window	Overlap context	Redundant, slow indexing	Q&A systems
Structure-aware	Respects formatting	Requires parsing logic	Mixed-format docs (PDFs, HTML)

Recommended Reading/Resources

- **Papers:** Lewis et al. (RAG), Gao et al. (Retrieval-Augmented Generation for Large Language Models)
 - **Libraries:** LangChain, LlamaIndex, Haystack, Unstructured, LiteLLM, Qdrant
 - **Eval Frameworks:** RAGAS, TruLens, DeepEval
 - **Your prior RAG training notes** (reuse your session material!)
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Labs Summary

Lab	Focus	Deliverable
Lab 1	Naive RAG	Working pipeline + 10 Q&A tests
Lab 2	Chunking experiments	Comparison table (3+ strategies)
Lab 3	Hybrid retrieval	BM25 + dense search code

Lab	Focus	Deliverable
Lab 4	Reranking	Cross-encoder integration
Lab 5	Evaluation	Metrics dashboard (precision, recall, NDCG)
Lab 6	Capstone	End-to-end RAG system + report

Success Criteria by Day 5

- ✓ Intern can explain RAG pipeline end-to-end
- ✓ Understands 5+ chunking strategies & tradeoffs
- ✓ Can implement hybrid retrieval + reranking
- ✓ Can build & evaluate a RAG system
- ✓ Can reason about when to use which RAG variant

Pro Tips

- **Day 1 afternoon:** Use your real data/docs—don't wait for synthetic datasets
- **Day 2:** Chunking is the hidden superpower; spend time here
- **Day 3-4:** Invest in evals early; most RAG bugs hide in retrieval quality
- **Day 5:** Leave time for optimization (faster indexing, cheaper embeddings, etc.)
- **Pair programming:** Do labs 2-5 together; catch bugs & learn faster